UK Train Rides Data Analysis

Project Proposal & Execution Plan

Project Title: UK Train Rides Analytics

Repository: NadaAtia688/DEPI-Project-

Team Lead: Nada Atia Abd Rabbo Mousa

Primary Tools: Power BI, Power Query, DAX

1. Executive Summary & Objectives

This project provides a comprehensive data analysis of UK train travel performance, revenue drivers, and passenger ride patterns. Utilizing Power BI, Power Query, and DAX, the team will transform raw train ride records into actionable business insights and predictive forecasting models.

Core Project Goals:

- Clean and structure raw UK train journey and transaction data.
- Formulate key business questions to support decision-making.
- Forecast ride volume and revenue for the upcoming month using DAX modeling.
- Analyze demand distribution across ticket classes and travel times.
- Deliver an interactive, user-friendly Power BI dashboard and final executive report.

2. Project Roadmap (4-Week Schedule)

Phase	Core Tasks	Key Deliverables
Week 1	Data Cleaning & Preprocessing using Power Query	Cleaned dataset & Data preprocessing notebook
Week 2	Formulate analysis questions, predict next month's rides/revenue & ticket class demand	Set of validated analysis questions & predictive models
Week 3	Build interactive Power BI Dashboard & Visualizations	Published Power BI Dashboard (.pbix)
Week 4	Final report preparation, slide deck creation & presentation	Final analytical report & presentation deck

3. Task Allocation Summary

To ensure clear ownership and efficient execution, project responsibilities have been directly mapped across team members according to the chronological project workflow:

Team Member	Assigned Task	Primary Responsibility
Doaa Ahmed	Data Cleaning	Data preprocessing, Power Query ETL, data transformation, and dataset staging.
Khalid Walid	DAX Modeling	Writing DAX measures, calculated columns, KPIs, and forecasting logic for rides & revenue.
Abdullah Al-Khazraji	Data Analytics	Trend analysis, statistical evaluation, ticket class demand modeling, and insight extraction.
Nada Atia (TL)	Dashboard & Supervision	Project oversight, timeline tracking, Power BI dashboard design, and UI/UX layout.
Zeinab Mohamed	Documentation & Reporting	Final project report, presentation slide deck preparation, and final project documentation.

4. Technical Stack Breakdown

Power Query

Extracts, transforms, and loads raw train ride datasets. Standardizes date formats, handles missing values, and creates relational schema.

DAX Engine

Calculates total revenue, ride counts, delay rates, and time-intelligence measures for predicting future month demand and daily projections.

Power BI

Delivers interactive dashboards, visual charts, and filtering slicers (by ticket class, route, time of day) for stakeholder presentation.

5. Closing & Sign-off

This simplified proposal outlines the complete scope, roadmap, and team responsibilities for the UK Train Rides Data Analysis initiative. The project team is prepared to commence Week 1 data cleaning immediately upon approval.

Questions or Feedback? Please contact Team Lead **Nada Atia** via the official project repository at `github.com/NadaAtia688/DEPI-Project-`.